Edu-tracker

Project Introduction

The Edu-tracker is a digital solution designed to improve communication between schools, teachers, parents, and learners regarding academic performance.

Currently, parents in many schools only receive information about their children's academic performance at specific times, such as at the end of a term when report cards are issued. This means that parents may not become aware of poor academic performance until several weeks or months after marks have been obtained.

The proposed system will allow teachers to capture and upload learners' marks electronically. Once marks have been recorded and made available, parents will be able to log into the system and view the academic performance of their children. This will provide parents with more timely access to academic information and enable them to identify areas where their children may require additional support.

The system aims to bridge the communication gap between the school and parents by making learner performance information more accessible. The solution will consist of a mobile application developed using Flutter, a backend developed using ASP.NET, and Supabase for data storage and related backend services.

The primary purpose of the project is to create a secure, user-friendly, and integrated platform that supports the monitoring of learner academic performance.

Stakeholder Profile and Current Process

Organisation:

The organisation involved in this project is a school environment consisting of school management, teachers, parents, and learners.

The school is responsible for managing learner academic information, including marks obtained from tests, assignments, examinations, and other assessments.

Stakeholders:

- Teachers

Teachers are responsible for assessing learners and recording their academic marks. They require an efficient way to capture and update learner marks.

- Parents or Guardians

Parents and guardians require access to information about their children's academic performance. They need to monitor marks and identify subjects in which their children may require additional assistance.

- System Administrator

The system administrator is responsible for managing user accounts, maintaining system information, and ensuring that the system operates correctly.

Current Process

The current process of communicating learner academic performance is largely delayed and may involve teachers recording marks manually or through internal school systems.

Parents generally receive formal information about their children's academic performance at the end of an academic term. As a result, there may be a significant delay between a learner receiving a mark and the parent becoming aware of it.

In the current process:

1. Teachers assess learners through tests, assignments, examinations, and other assessments.

2. Teachers record learner marks.

3. Marks may remain accessible only to teachers or the school until reports are issued.

4. Parents wait until the end of a term or until communication is provided by the school.

5. Parents may only become aware of academic problems after a considerable period of time.

This process limits the ability of parents to monitor academic progress continuously.

Problem Statement

The current process of communicating learner academic performance to parents is often delayed. Parents may only receive information about their children's marks at the end of an academic term. This creates a communication gap between teachers, schools, and parents.

As a result, parents may not identify poor academic performance early enough to provide the necessary support. Learners who are struggling in specific subjects may therefore continue experiencing difficulties before their parents become aware of the problem.

Teachers may also require a more efficient method of capturing and managing learner marks. A lack of an accessible system can make it difficult to provide parents with timely academic information.

The main problem is therefore the lack of a centralised and timely system that allows teachers to capture learner marks and allows parents to monitor their children's academic performance.

The system will address this problem by providing a platform through which authorised teachers can upload learner marks and authorised parents can view the academic performance of their children.

Project Aim and Objectives

Project Aim:

The aim of this project is to develop a secure Edu-tracker that enables teachers to manage learner marks and allows parents to access and monitor their children's academic performance.

Project Objectives:

The objectives of the project are to:

1. Develop a system that allows authorised users to register and log in securely.
2. Allow teachers to capture and upload learner marks.
3. Allow teachers to update or correct marks where necessary.
4. Allow parents to view the academic marks of their linked children.
5. Display learner performance according to subjects and assessments.
6. Provide parents with timely access to updated academic information.
7. Improve communication between teachers, schools, and parents.
8. Protect learner academic information through secure authentication and authorisation.
9. Develop a user-friendly mobile application using Flutter.
10. Develop backend services using ASP.NET.
11. Store and manage system data using Supabase.
12. Ensure that the system components operate together as an integrated solution.

Project Scope

In-Scope Features:

The following features are included in the scope of the proposed system:

1. User Authentication

The system will allow authorised users to register or log in using secure credentials.

2. User Roles

The system will support different user roles, including:

- Teacher
- Parent or guardian
- System administrator

Each user role will have access only to features relevant to their responsibilities.

Learner Management:

The system will store learner information required for academic monitoring.

Parent-Learner Association:

Parents or guardians will be linked to their children so that they can only access the academic information of authorised learners.

Subject Management:

The system will store subjects associated with learners and their academic performance.

Marks Management:

Teachers will be able to:

- Select a learner.
- Select a subject.
- Capture an assessment mark.
- Specify the assessment type.
- Save the mark.
- Update a mark where necessary.

Parent Dashboard:

Parents will be able to view:

- Their children's subjects.
- Assessment marks.
- Academic performance.
- Areas where academic improvement may be required.

Performance Monitoring:

The system will present learner marks in a clear format to assist parents in monitoring academic progress.

Deliverables:

The project deliverables will include:

1. A Flutter mobile application.

2. An ASP.NET backend or API.

3. A Supabase database configuration.

4. User authentication functionality.

5. Teacher marks management functionality.

6. Parent learner-progress viewing functionality.

7. Database documentation.

8. System requirements documentation.

9. Testing documentation.

10. A final integrated prototype or working system.

Acceptance Criteria:

The system will be considered acceptable if:

- Users can successfully authenticate.
- Teachers can capture learner marks.
- Captured marks are stored successfully.
- Parents can view only the marks of their authorised children.
- Updated marks are reflected correctly in the system.
- The Flutter application communicates successfully with the ASP.NET backend.
- The ASP.NET backend communicates successfully with Supabase.
- Unauthorised users cannot access protected academic information.
- The main system features operate without critical errors.

Exclusions:

The following features are currently outside the scope of the project:

- Full school financial management.
- School fee payments.
- Learner attendance management.
- Online classroom functionality.
- Video lessons.
- Automated grading using artificial intelligence.
- Complete school administration and human resources management.
- Public access to learner academic information.

Constraints

Possible project constraints include:

- Limited development time.
- Limited access to real school data due to privacy concerns.
- Internet connectivity requirements.
- Limited testing with actual users.
- Data protection requirements.
- Limited resources for hosting and deployment.

- The need to integrate multiple technologies.

Assumptions

The project assumes that:

- Teachers will have access to a device capable of using the system.
- Parents will have access to a compatible smartphone.
- Users will have internet access when accessing online functionality.
- The school will provide accurate learner information.
- Parents will only be linked to their authorised children.
- Teachers will enter accurate marks.

Stakeholder Needs

Teacher Needs

Teachers need the system to:

- Capture learner marks efficiently.
- Update marks when corrections are required.
- View learners associated with their classes.
- Manage academic information accurately.
- Reduce unnecessary delays in communicating marks.

Parent Needs

Parents need the system to:

- Access their children's academic information easily.
- View marks without waiting until the end of the term.
- Identify subjects in which their children may be struggling.
- Monitor changes in academic performance.
- Access information securely.

School Management Need

School management needs:

- Improved communication with parents.
- Accurate management of academic information.

- Controlled access to learner information.
- A reliable and secure system.

Functional Requirements

FR1: User Authentication

The system shall allow authorised users to register and log in using valid credentials.

Priority: High

FR2: Role-Based Access

The system shall provide different access levels according to the user's role.

Teachers shall access teacher-related functions.

Parents shall only access information relating to their authorised children.

Administrators shall have access to authorised administrative functions.

Priority: High

FR3: Teacher Captures Marks

The system shall allow teachers to capture learner marks.

The teacher shall be able to enter:

- Learner information.
- Subject.
- Assessment type.
- Assessment date.
- Mark obtained

Priority: High

FR4: Update Learner Marks

The system shall allow authorised teachers to update learner marks when corrections are required.

Priority: High

FR5: View Learner Marks

The system shall allow parents to view the academic marks of their authorised children.

Priority: High

FR6: View Performance by Subject

The system shall display learner marks according to subjects.

Parents shall be able to identify the subjects in which learners perform well or may require improvement.

Priority: High

FR7: Parent-Learner Linkin

The system shall link a parent or guardian to the relevant learner or learners.

Parents shall not be able to view information relating to learners who are not linked to their accounts.

Priority: High

FR8: Assessment History

The system shall maintain a history of recorded assessment marks.

Priority: Medium

FR9: Search and Selection

The system shall allow authorised teachers to search for or select learners when capturing marks.

Priority: Medium

FR10: Data Storage

The system shall store relevant user, learner, subject, and academic mark information securely

Priority: High

FR11: Error Message

The system shall provide meaningful error messages when an operation cannot be completed.

Examples include:

- Invalid login details.
- Missing required information.
- Invalid marks.
- Unauthorised access.

Priority: High

Non-Functional Requirements

Usability

The system shall provide a simple and user-friendly interface.

Users should be able to understand the main functions without extensive training.

The application shall use clear labels and understandable navigation.

Performance

The system should retrieve and display learner academic information within a reasonable time under normal network conditions

Marks should be saved without unnecessary delays.

Security

The system shall require user authentication before protected information can be accessed.

Users shall only access information authorised for their role.

Parents shall only access the academic information of their linked children.

Sensitive information shall be protected during storage and transmission.

Availability

The system should be available to authorised users whenever the supporting services and internet connection are available.

Reliability

The system shall store marks accurately and maintain data consistency.

The system should prevent invalid data from being saved.

Maintainability

The system should be designed in a modular manner to allow future changes and improvements.

The Flutter application, ASP.NET backend, and Supabase database should have clear responsibilities.

Compatibility

The mobile application shall support the Android platform

Future versions may support additional platforms.

Data Privacy

The system shall protect learner and parent information

Access to academic records shall be restricted to authorised users.

Integration Requirement

The proposed system consists of three major technologies that must operate as an integrated solution.

Flutter Mobile Application

Flutter will be used to develop the mobile application.

The Flutter application will provide the user interface through which teachers and parents interact with the system.

The application will be responsible for:

- User login.
- Displaying dashboards.
- Capturing marks.
- Viewing learner performance.
- Sending requests to the ASP.NET backend.
- Displaying responses and error messages.

ASP.NET Backend

ASP.NET will provide the backend services and application programming interfaces.

The ASP.NET backend will be responsible for:

- Processing requests from the Flutter application.
- Applying business rules.
- Validating user input.
- Managing authorised access.
- Communicating with the Supabase database.
- Returning appropriate responses to the mobile application.

The Flutter application will communicate with the ASP.NET backend through API requests.

Supabase

Supabase will be used to store and manage system data.

The database will contain information relating to:

- Users
- Parents
- Teachers
- Learners
- Subjects
- Parent-learner relationships
- Academic assessments
- Learner marks

Supabase will provide the required data services for the integrated solution.

Integration Flow

The expected flow of information will be:

Flutter Mobile Application → ASP.NET API → Supabase Database

For example:

1. A teacher enters a learner's mark in the Flutter application.

2. The Flutter application sends the mark information to the ASP.NET API.

3. The ASP.NET API validates the information.

4. The ASP.NET API stores the information in Supabase.

5. The updated information becomes available to authorised users.

6. A parent accesses the Flutter application.

7. The application requests the authorised learner's marks through the ASP.NET API.

8. The ASP.NET API retrieves the required information from Supabase.

9. The learner's marks are displayed to the parent.

Requirements Change Log

Change Number	Original Requirement	Updated Requirement	Reason for Change
RC01	Broad school communication system	Focus on learner academic marks and performance	To keep the project manageable
RC02	Immediate access to all school information	Parents access academic information only	To protect privacy and reduce scope
RC03	General user access	Role-based access for teachers and parents	To improve security
RC04	Teachers and parents use separate systems	One integrated system with different user roles	To improve communication and maintainability
RC05	Direct mobile-to-database communication	Flutter application communicates through an ASP.NET backend	To support business logic, validation, and controlled integration
RC06	Full school management functionality	Focus on academic performance monitoring	To ensure the project is realistic within the available time
RC07	Access to all learner information	Parents access only linked learners	To protect confidential information

Refined Project Scope

After considering feasibility, risks, project constraints, and the available development time, the project scope has been refined.

The system will focus on the core functionality required to solve the identified problem.

The refined system will include:

- User authentication
- Teacher access
- Parent access
- Role-based access control
- Learner information management
- Parent-to-learner linking

- Subject information
- Teacher marks capture
- Marks updates
- Parent viewing of learner marks
- Basic performance monitoring
- Integration between Flutter, ASP.NET, and Supabase

The following features will not be included in the initial version:

Online payment systems

- Attendance tracking
- Full learner registration processes
- Artificial intelligence-based academic predictions
- Automated notifications for every academic event
- Video conferencing
- Online learning functionality
- Complete school administration

These features may be considered for future development after the core learner progress monitoring functionality has been successfully implemented.

Feasibility Study

This feasibility study evaluates the Edu-tracker proposed for Mangaung Primary School. It follows the six required feasibility areas and is based on the confirmed stakeholder problem, the SRS, and the Project Risk Register.

Review of the Current Project

Question	Group Finding
What is the main problem being addressed?	Parents only find out about their children's marks at the end of term, often because results are lost, not relayed, or communicated informally by the learner. This delays parental awareness of academic difficulty.
What are the main Flutter functions?	User login for teachers and parents; teacher screens to capture and update marks; parent dashboard to view linked learners' subjects, marks and performance.
What are the main ASP.NET functions?	Processing requests from the Flutter app, applying business rules and input validation, enforcing role-based access, and communicating with the Supabase database.
What information will Supabase manage?	Users, teachers, parents, learners, subjects, parent-learner relationships, assessments and learner marks.

What are the stakeholder's expectations?	Teachers want an efficient, low-effort way to capture and correct marks. Parents want timely, secure access to only their own children's academic information (SRS Stakeholder Needs, §6).
What skills does the group currently have?	Flutter UI development and basic ASP.NET API development and relational database design. Secure API authentication and Supabase row-level security are less developed within the group.
What does the Microsoft Project schedule show?	Remaining phases cover system design, Flutter development, ASP.NET development, Supabase integration, testing and corrections, and final presentation preparation.
What are the main project constraints?	Limited development time, limited access to real school data, dependency on internet access, limited real-user testing, data protection requirements, and integrating three separate technologies (SRS §5.5).
Which functions appear most difficult?	Role-based access control enforced consistently across Flutter, ASP.NET and Supabase, and correctly linking each parent to only their own children (FR2, FR7).
Which functions require skills the group still needs to develop?	Supabase row-level security policies and secure authentication/authorisation in the ASP.NET API.
Which functions may require more time than is available?	End-to-end integration testing across all three layers, and structured usability testing with an actual teacher.

Technical Feasibility

Question	Group Finding
Does the group have the required Flutter skills?	Partially — some members have prior Flutter experience; core UI screens are within reach.
Does the group have the required ASP.NET skills?	Partially — basic API development is understood; secure role-based access needs further learning.
Can the group work with Supabase?	Yes — Supabase's free tier and PostgreSQL basics are understood by the group.
Can Flutter and ASP.NET connect to the same Supabase backend?	Yes — Flutter will call the ASP.NET API, which alone communicates with Supabase, keeping data access centralised.
Are suitable development computers available?	Yes — group members have their own laptops.
Are suitable test devices available?	Partially — Android phones and emulators are available; testing on real low-end devices is limited.
Is reliable internet access available?	Yes, for development; parent-side internet access at the school cannot be fully confirmed.
Which technical skills still need to be learned?	Supabase row-level security and secure ASP.NET authentication/authorisation.
Which functions may be too technically difficult?	None are considered infeasible; the RBAC/integration work is the main challenge but is achievable within scope.

Operational Feasibility

Question	Group Finding
Will the prototype address the stakeholder's problem?	Yes — it replaces the current delayed, informal process with real-time, role-restricted access to marks.
Who will use the mobile application?	Teachers (marks capture/update) and parents (viewing linked learners' marks).
Who will use the web application?	The ASP.NET component is primarily a backend/API in the current design; whether a separate web interface is needed (e.g., for administrators) still needs to be confirmed with the stakeholder.
Do the intended users have suitable devices?	Teachers likely have access to school computers; parents are assumed to have a compatible smartphone, which is not guaranteed for every household.
Do users have suitable internet access?	Assumed but not fully confirmed, particularly for parents (SRS assumptions, §5.6).
Will the proposed solution fit the organisation's current activities?	Yes — it digitises an existing manual process (recording and distributing marks) rather than replacing it entirely.
Will users require guidance or training?	Minimal for parents given a simple dashboard; teachers will need brief onboarding to the marks-capture workflow.
Are any proposed functions unsuitable for the intended users?	Automated notifications for every academic event were excluded to avoid overwhelming users (SRS refined scope, §11).

Decision: Feasible with changes — confirm parents' device and internet access with the stakeholder and provide short teacher onboarding.

Economic Feasibility

Resource	Available?	Action Required
Internet access or data	Available (group)	None — parent-side data costs to be noted as a future consideration.
Transport to the stakeholder	Available	Schedule stakeholder visits/calls in advance.
Development computers	Available	None.
Test devices	Partially available	Borrow a low-end Android device for accessibility testing.
Required software	Available (free)	None — Flutter SDK, .NET SDK, Supabase free tier.
Supabase resources	Available (free tier)	Monitor free-tier usage limits.
Presentation resources	Available	None — university facilities.
Other project resources	None currently required	N/A

Decision: Feasible — all required resources are available at little or no cost.

Schedule Feasibility

Question	Group Finding
Is enough time available for system design?	Yes — allocated early in the Microsoft Project schedule.
Is enough time available for Flutter development?	Yes, though scheduled tightly alongside integration work.
Is enough time available for ASP.NET development?	Yes — scheduled in parallel with Flutter development.
Is enough time available for Supabase integration?	Allocated, but at risk if the RBAC/security learning curve is steeper than expected.
Is enough time available for testing and corrections?	Allocated, with limited buffer for unexpected issues.
Is enough time available for stakeholder feedback?	Yes — feedback points are scheduled at key milestones.
Are too many functions currently included?	No — the scope was already trimmed via the Requirements Change Log (RC01–RC07).
Which functions may need to be simplified, postponed or removed?	Assessment history (FR8) and search/selection convenience features (FR9) can be treated as stretch goals if time runs short.

Decision: Feasible with changes — treat FR8 and FR9 as stretch goals to protect the schedule buffer.

Legal Feasibility

Question	Group Finding
Will the prototype collect personal information?	Yes — learner names, marks, subjects, and parent-learner relationships.
Is all the information necessary?	Yes — scope was deliberately reduced to only what is needed for marks monitoring (Change Log RC02, RC07).
Who should be allowed to access the information?	Teachers (their own learners), parents (only their linked children), and administrators (system-level access), enforced through role-based access control.
Will test data be used during development?	Yes — dummy/fictional learner data will be used; real learner records will not be used for development or testing.
How will passwords and private keys be protected?	Stored as environment variables/secrets, never committed to the GitHub repository.
Does the project involve confidential stakeholder information?	Yes — real school and process details were discussed with the stakeholder but will not be published.
Is private stakeholder information being kept out of GitHub?	Yes — only documentation and non-identifying/sample data will be pushed to the repository.

Decision: Feasible with changes — strictly enforce dummy data during development and confirm no real learner data reaches GitHub or submitted evidence.

Ethical Feasibility

Question	Group Finding
Does the stakeholder understand the purpose of the project?	Yes — confirmed through stakeholder engagement and evidence gathered during Units 1–2.
Is unnecessary personal information being collected?	No — data collected is limited to what is needed for marks monitoring.
Could the proposed solution negatively affect intended users?	Possibly — real-time visibility could increase pressure on a struggling learner before they have a chance to improve; mitigated by presenting marks with context rather than in isolation.
Is information being used only for its intended purpose?	Yes — access is restricted to informing the authorised parent; no ranking or public comparison of learners.
Does the proposed prototype accurately reflect stakeholder needs?	Yes — it is aligned to the confirmed problem statement, and stakeholder needs (SRS §3, §6).
Are users being treated fairly and responsibly?	Yes — access controls and scope exclusions (no public access, no ranking) support fair treatment.

Decision: Feasible.

Feasibility Summary

Feasibility Area	Decision	Main Reason	Change Required
Technical	Feasible with changes	Stack matches group skills, but RBAC/security spans two backend layers and is not yet fully mastered.	Learn Supabase RLS and ASP.NET secure auth early.
Operational	Feasible with changes	Solves the confirmed problem, but parent device/internet access is not fully confirmed.	Confirm device/internet access with stakeholder; short teacher onboarding.
Economic	Feasible	All required resources are available at little or no cost.	Nonmajor.
Schedule	Feasible with changes	Scope already trimmed, but integration work carries schedule risk.	Treat FR8/FR9 as stretch goals.
Legal	Feasible with changes	Involves personal data on minors under POPIA.	Use dummy data only; keep real data out of GitHub.
Ethical	Feasible	Data use is restricted to informing the authorised parent; no misuse identified.	Nonmajor.

Overall Feasibility Decision

Overall decision: The project is feasible with changes.

The proposed project is feasible with changes. The main strengths of the project are a real, confirmed stakeholder problem, a technology stack that matches the group's combined skills, and a project scope that has already been trimmed to fit the semester through the Requirements Change Log. The main limitations identified are the group's limited experience with role-based access control across two backend layers, and unconfirmed assumptions about parents' device and internet access. To ensure the project can be completed within the semester, the group will prioritise learning secure authentication and access control early, use only dummy/test data during development, confirm device access assumptions with the stakeholder, and treat lower-priority functions — assessment history and advanced search — as stretch goals that may be postponed if time runs short.

Project risk register

The implementation of a real-time learner marks tracking system carries potential risks related to data privacy, system availability, teacher adoption, data entry errors, and device accessibility that could hinder parents' ability to monitor academic progress effectively.

Risk ID	Risk Category	Probability	Impact	Score	Contingency action	Risk owner	Current risk status	Mitigation Action
R01	Threats related to the system's security and user privacy	High	High	9	Use role-based access control patches and lock out compromised user accounts.	Mihlali Mtyali	Open	Enforce strong authentication rules and restrict access to key system components.
R02	How the system can struggle to get user adoption	High	High	9	Provide users with training and simplify the system's interface.	Tlamelo Mothibi	Monitoring	Involve teachers early in user testing and design a streamlined architecture.
R03	The credibility of the platform is tied to the integrity of the data	Medium	High	6	Allow teachers to send correction requests and notify parents of mark revisions.	Nonzuzo Shabalala	Open	Add input validation checks and confirmation prompts before publishing.
R04	The system's ongoing availability for the user base	Medium	Medium	4	Provide offline mark entry caching that syncs automatically	Sachin Van Aarde	Open	Hosting the system on a scalable cloud architecture

					once back online.			with database backup.
R05	Accessibility to various devices for a diverse user base	Medium	Medium	4	Develop a lightweight mobile view optimized for low-bandwidth cellular data.	Nkagisang Marate	Open	Implement responsive web design so parents can log in easily on low-end smartphones.
R06	Technical challenges that are related to integration for 2 types of applications	Medium	High	6	Establish an integration test environment early; define clear API contracts between the Flutter app, ASP.NET layer, and Supabase before development begins.	Lerato Theoha	Open	Adopt consistent API versioning and run automated integration tests whenever any of the three components (Flutter, ASP.NET, Supabase) is updated.
R07	Scheduling all the required work into the available time	High	High	9	Prioritise core features (authentication, marks capture, marks viewing).	Tlamelo Mothibi	Open	Use regular milestone reviews to track progress against the project timeline.
R08	Ensuring that the skills needed are available in the team	Medium	Medium	4	Allocate time for team members to execute what is expected based on their skill level.	Kabelo Ramolise	Open	Practice and revise Flutter, ASP.NET, and Supabase training and skills to ensure readiness.
R09	The degree to which implementation depends on vendor's	Low	Medium	2	Monitor Supabase service status and plan for temporary local caching or fallback handling during outages.	Ntsoaki Ramohloki	Monitoring	Review Supabase's service level terms and usage limits also keep an alternative provider that could be substituted if required.

Three highest project risks

R01: Data Privacy Breach (Security Risk): Unauthorized access to confidential student marks could violate learner privacy rights and break trust between parents and the school.

R02: Teacher Resistance & Low Adoption (Operational Risk): If teachers find real-time mark uploading burdensome, data will not be updated promptly, rendering the system ineffective for parents.

R03: Scheduling risk: Inappropriately scheduling tasks may adversely affect the team's ability to complete the project in the allotted time, thus affecting prospects of success.

Conclusion

The Edu-tracker is intended to address the communication gap between schools and parents regarding learner academic performance

The proposed solution will allow teachers to capture learner marks and allow parents to access their children's academic performance in a timelier manner. This can assist parents in identifying areas where learners require additional academic support.

The system will use Flutter for the mobile application, ASP.NET for backend services, and Supabase for data management. The integration of these technologies will provide a structured solution for managing and accessing learner academic information.

By focusing on the core requirements of marks management, secure access, parent monitoring, and system integration, the project provides a realistic and achievable scope for the next stages of system design and development.

GitHub Link

- GitHub Repo

Supabase Link

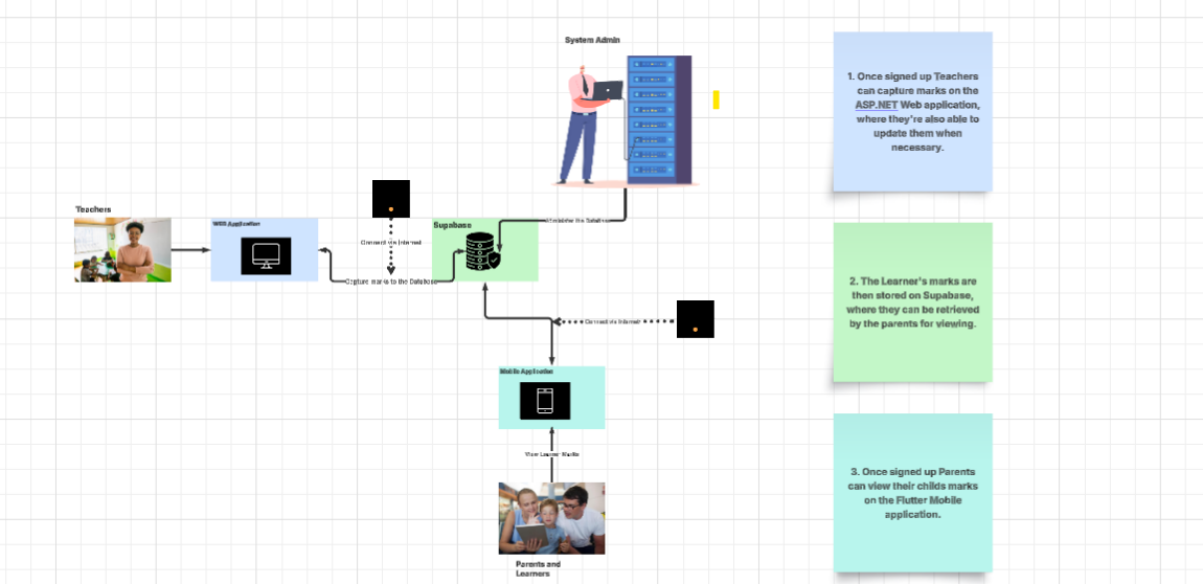
- Supabase Database

Dropbox Link

- Dropbox Document Storage

Phase 2

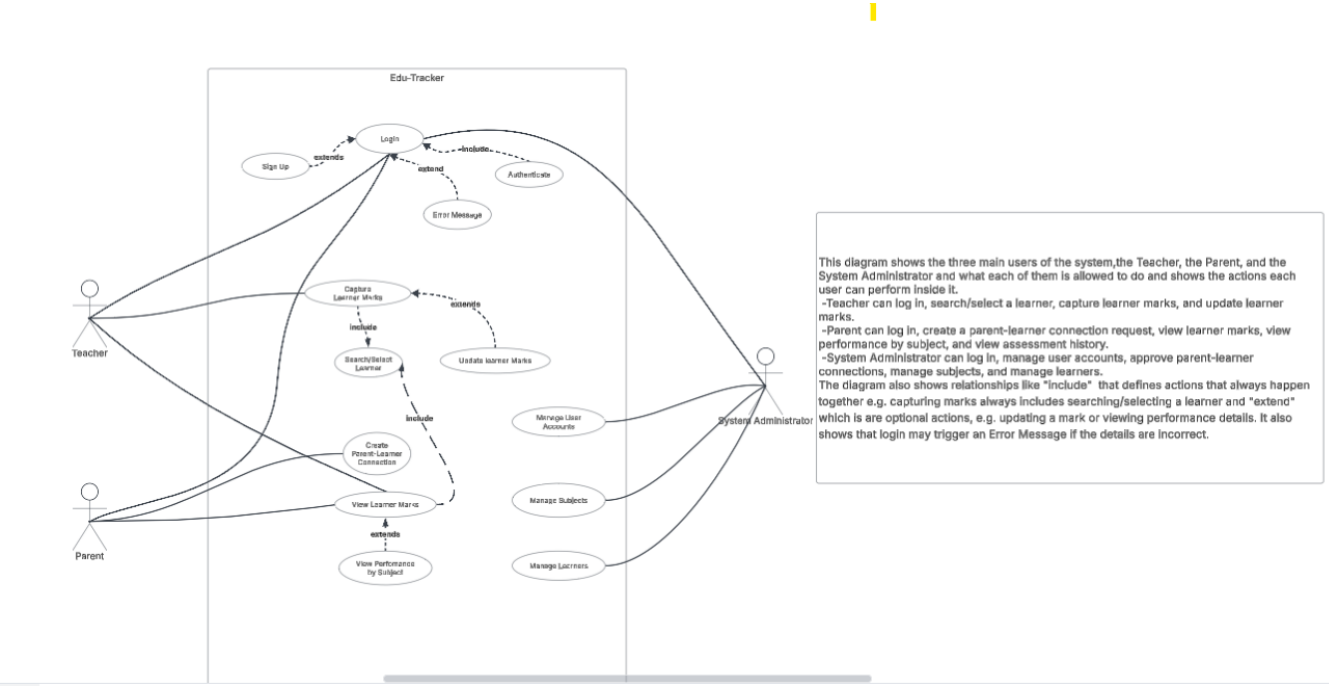
SYSTEM ARCHITECTURE DESIGN



System Architecture Design Link

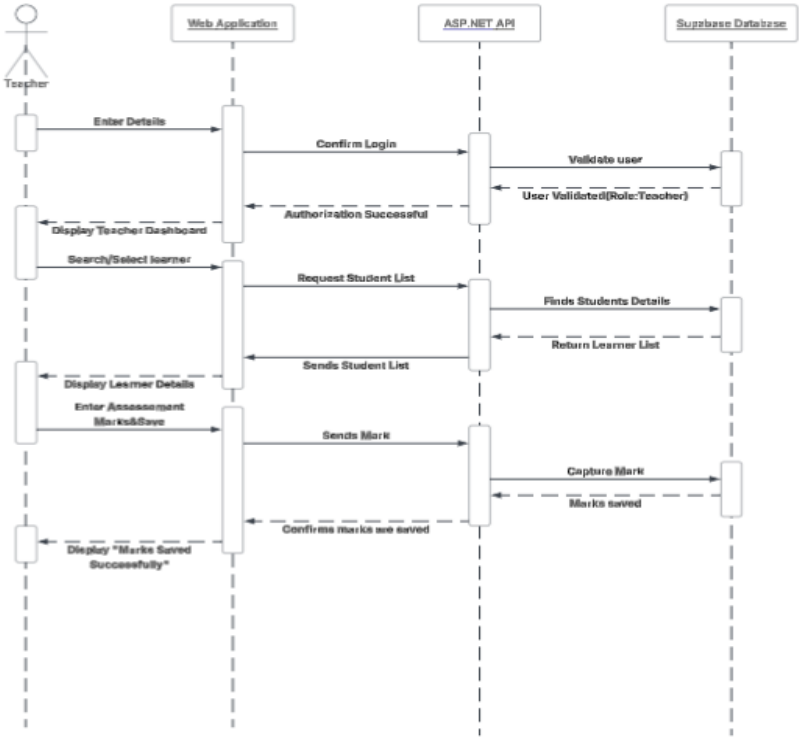
UML DESIGNS

Use Case Diagram

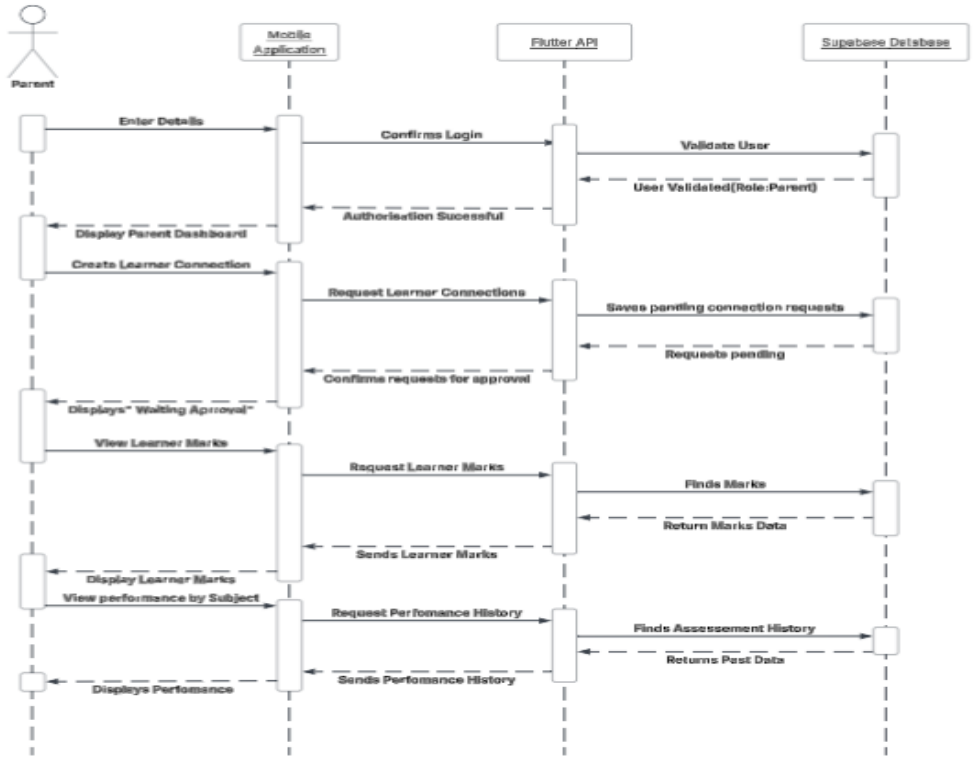


UML Design Link

Sequence Diagram Teacher



Sequence Diagram Parent



Sequence Diagram Link

Web App Interface

Web app interface link

Mobile App Interface

Mobile app Interface Link

System Design Update

Design Artifact	Changes Made	Reasoning

Use Case Diagram	Added “includes” and “extends”.	To clarify the link between the use cases.
Entity Relationship Diagram	Refined the “Marks” and “Assessment” entities.	To remove redundancy and improve efficiency.
Mobile App Interface	Added back/return buttons.	They are part of the normal operation of the app.
Web App Interface	Change colours to align with the school’s colours.	Makes it more user friendly and specific.